

Simulation of food drying: FEM analysis and experimental validation

Stefano Curcio *, Maria Aversa, Vincenza Calabrò, Gabriele Iorio

Department of Engineering Modeling, University of Calabria, Ponte P. Bucci – Cubo 39/c, 87030 Rende, CS, Italy

Received 26 October 2007; received in revised form 11 January 2008; accepted 14 January 2008

Available online 31 January 2008

Abstract

The aim of the present work is the formulation of a theoretical model describing the simultaneous transfer of momentum, heat and mass occurring in a convective drier where hot dry air flows under turbulent conditions around a food sample. The proposed model does not rely on the specification of interfacial heat and mass transfer coefficients and, therefore, represents a general tool capable of describing the behavior of real driers over a wide range of process and fluid-dynamic conditions. The system of non-linear unsteady-state partial differential equations modeling the behavior of a cylindrical-shaped vegetable sample in a drier, has been solved by using finite elements method. It has been observed that air characteristics influence drying performance only when external resistance to mass transfer is the rate controlling step. An experimental study was undertaken which shows very good agreement between model predictions and experimental results.

© 2008 Elsevier Ltd. All rights reserved.

Keywords: Food drying; Transport phenomena; Finite elements method; Process modeling

1. Introduction

When warm dry air flows around a cold moist food sample, simultaneous heat and mass (water) transfer occurs, leading to, both, a decrease in food water content and an increase in its temperature. The above effects enhance food preservation, since microbial spoilage is generally promoted by low temperature and high moisture content. Heat and mass transfer rates depend on both temperature and concentration differences, but also on the air velocity field which strongly influences the transfer rate at food–air interfaces and, therefore, has to be properly evaluated. An exhaustive analysis of all the complex transport phenomena involved in drying process has often been regarded as being too onerous and time consuming for practical purposes. For this reason, many simplified approaches have been proposed, and are widely used by industrial drier

designers (Mujumdar Arun, 2006). These approaches are based either on simplifying hypotheses, which may not be applicable in practice, or on the use of semi-empirical correlations for estimating the heat and water fluxes at food–air interfaces (Saravacos and Maroulis, 2001).

Hernández et al. (2000) assumed fruit drying as an isothermal process occurring at a fixed air temperature; this simplification restricted the analysis to mass transfer only. Wu and Irudayaraj (1996) experimentally verified that drying is an isothermal process only if Biot number is very low. When Biot number is significantly greater than unity, the internal transport resistances are not negligible. Wang and Brennan (1995) developed a one-dimensional model for the simultaneous heat and mass transfer within potato slices. The hypothesis of one-dimensional transport was experimentally verified in the same study and also adopted by other authors (Kalbasi and Mehraban, 2000; Rovedo et al., 1995; Migliori et al., 2005). In a recent paper, Datta (2007 Part I) showed the different approaches that have to be used to model heat and mass transfer in food drying process. In some cases, the size of the pores as well as vapor generation within the sample have to be taken into consideration, whereas in some others cases, they can be

* Corresponding author. Tel.: +39 0984 496 711/670/703/709; fax: +39 0984 496671.

E-mail addresses: stefano.curcio@unical.it (S. Curcio), maria.aversa@unical.it (M. Aversa), vincenza.calabro@unical.it (V. Calabrò), gabriele.iorio@unical.it (G. Iorio).

Nomenclature

C	water concentration in food, mol/m ³	u_0	air velocity at the drier inlet, m/s
C^+	scalar variable, –	U_r	relative humidity of air, –
C_2	water concentration in air, mol/m ³	$\bar{X}$	average moisture content-dry basis, kg water/kg dry solid
C_{2i}	water concentration in gaseous phase at the food/air interface, mol/m ³	X_e	equilibrium moisture content on a dry basis, kg water/kg dry solid
C_{pa}	air specific heat, J/(kg K)	x_w	molar fraction of water in the food, –
C_{ps}	food specific heat, J/(kg K)	y_w	molar fraction of water in air, –
c_μ	model parameter, –	z	spatial coordinate, m
$c_{1\varepsilon}$	model parameter, –	Greek Symbols	
$c_{2\varepsilon}$	model parameter, –	γ_w	activity coefficient of water (liquid phase), –
D_a	diffusion coefficient of water in air, m ² /s	δ_w	distance from the wall, m
D_{eff}	effective diffusion coefficient of water in food, m ² /s	δ_w^+	dimensionless distance from the wall, –
f_w^0	fugacity of water (liquid phase), Pa	ε	turbulent energy dissipation rate, m ² /s ³
I	identity matrix, –	η_a	air viscosity, Pa*s
k	turbulent kinetic energy, m ² /s ²	η_t	air turbulent viscosity, Pa*s
k_a	air thermal conductivity, W/(m K)	λ	water latent heat of vaporization, J/kg
k_c	Karman's constant, –	ρ_a	air density, kg/m ³
k_{eff}	effective thermal conductivity of food, W/(m K)	ρ_s	food density, kg/m ³
$\underline{n}$	unity vector normal to the surface, –	σ_ε	model parameter, –
p	pressure within the drying chamber, Pa	σ_κ	model parameter, –
P_w^{sat}	vapor pressure of water, Pa	ϕ	dimensionless average moisture content of food, –
R	gas constant, J/(mol K)	$\hat{\phi}_w$	fugacity coefficient of water-vapor in solution, –
r	spatial coordinate, m	ϕ_w^{sat}	fugacity coefficient of water-vapor at saturation conditions, –
T	food temperature, K	subscripts	
t	time, s	0	initial condition ($t = 0$)
T_2	air temperature, K	atm	atmospheric conditions
T_{air}	air temperature at the drier inlet, K	db	on a dry basis
$\bar{T}_b$	average temperature at the food surface, K	wb	on a wet basis
U	moisture content on a wet basis, kg water/kg wet solid		
$\bar{U}$	average moisture content on a wet basis, kg water/kg wet solid		
$\underline{u}$	velocity vector, m/s		

neglected. A transport model describing the simultaneous two-dimensional heat and moisture transfer was formulated to analyze the influence of some of the most important operating variables on the drying rate of carrots (Aversa et al., 2007). The authors solved the system of unsteady-state partial differential equations, modeling the transport phenomena occurring within the food, by means of the finite elements method. The dependence of physical and transport properties on food temperature and moisture content has been investigated by several authors (Datta, 2007 Part II; Panagiotou et al., 2004; Lewicki, 2004; Saravacos and Maroulis, 2001). The large number of available studies dealing with drying process suggests, however, that more general and versatile mathematical models must be formulated. These models can potentially overcome some problems regarding, for instance, the use of the most appropriate semi-empirical correlations for evaluating heat and mass transfer coefficients at the food/air interfaces.

Kondjoyan and Boisson (1997) and Verboven et al. (1997), in fact, found that even small errors in the estimation of transfer coefficients could lead to large deviations between estimated and real values of temperature and moisture content, leading to inappropriate equipment design or severe processing problems. It is therefore desirable to formulate a transport model accounting for the simultaneous transfer of momentum, heat and mass in air as well as within the food. Moreover, with a proper set of boundary conditions expressing the continuity of both heat and mass fluxes at the food/air interfaces, it is possible to estimate the actual transport rates without resorting to any empirical correlation. Such a model is therefore capable of predicting the drying behavior of foods available in all shapes, and over a very wide range of operating and fluid-dynamic conditions. The present work is intended to fill such a gap in modeling the drying of foods. The main objective of this paper is to formulate an accurate transport

model analyzing the simultaneous transfer of momentum (for air only), and of heat and mass (for both air and food) occurring in a convective drier where hot dry air flows, under turbulent conditions, around a cold wet food sample. The model can be used to study the time evolution of some characteristic parameters, namely the moisture content of food and its temperature, expressed as functions of the operating conditions, i.e. inlet velocity, relative humidity and temperature of the air. The model proposed minimizes expensive pilot scale tests and satisfactorily indicates the characteristics and safety of dried products.

2. Theoretical background

In a convective drier, two different transport mechanisms simultaneously occur: heat is transferred from air to the material; water is transported from the core of the material to its surface and, eventually, to air. The rates of heat and mass transfer certainly depend on both temperature and concentration differences, but also on air velocity field which any model has to properly predict. In the following, the main physical assumptions and the mathematical equations used to model the unsteady-state behavior of a convective drier will be discussed. It is assumed that drying air is continuously supplied to the oven inlet section and flows around a cylindrical food sample in the axial direction, parallel to its length (Fig. 1). Heat and mass transfer resistance are assumed negligible across the mesh on which the food is placed (Thorvaldsson and Janestad, 1999; Viollaz and Rovedo, 2002). The system under investigation is symmetric; and only half of the original domain will henceforth be considered. Moreover, variation occurring in the θ direction (i.e. angular coordinate) will be neglected, thus restricting the analysis to a 2D geometry. Each dependent variable is, therefore, function of the radial and axial coordinates, r and z , respectively, and of time, t . The drier under investigation has a length of 25 cm and a radius of 10 cm. The food sample, assumed

to be a half cylinder 6 cm long and 5 mm radius, is placed 8 cm far from the drier inlet.

The food sample is a multiphase hygroscopic porous medium; yet it is assumed to be a fictitious continuum. The contribution of convection to the transport equations written for the food sample has been neglected, assuming weak internal evaporation. Also the transport of water vapor by diffusion within the dehydrated material towards the external food surface, has not been considered since this mechanism is only significant in the case of highly porous media, whereas it can be neglected in the case of vegetables since typical void fraction values are less than 0.3 (May and Perré, 2002). Mass transfer in the product, therefore, occurs only by diffusion; and heat transfer, only by conduction. Shrinkage effects were assumed to be negligible in the range of moisture content considered here. The proposed model accounts for the variation of food physical and transport properties which are dependent on the local values of temperature and moisture content (Aversa et al., 2007).

In air, heat transfer occurs by convection and conduction, whereas the water is transferred by convection and diffusion. The convective contributions to heat and mass transfer are related to air circulation, and the velocity field is to be determined by solving the non-isothermal momentum transport equations for turbulent flow, coupled with the continuity equation. It is assumed that evaporation/condensation only occur at the air–food interface where a proper set of boundary conditions has to be specified. These boundary conditions substantially express the continuity of temperatures and heat and mass fluxes; thermodynamic equilibrium is assumed to prevail between the water concentration in air and the moisture content at the food–air interface. Physical and transport properties of air have been also expressed in terms of the local values of temperature and water content.

Following on from the Fick's law, the unsteady-state mass transfer equation for the movement of water within the food sample is (Bird et al., 1960; Welty et al., 2001)

$$\partial C / \partial t + \nabla \cdot (-D_{\text{eff}} \nabla C) = 0 \quad (1)$$

where C is the water concentration in food and D_{eff} is the effective diffusion coefficient of water in the food.

According to Fourier's law, the energy balance in the food material leads to the unsteady-state heat transfer equation (Bird et al., 1960; Welty et al., 2001):

$$\rho_s C_{ps} \partial T / \partial t - \nabla \cdot (k_{\text{eff}} \nabla T) = 0 \quad (2)$$

where T is temperature; ρ_s is the density of the food sample; C_{ps} its specific heat and k_{eff} is the effective thermal conductivity. The subscript eff refers to food transport properties evaluated as effective, i.e. accounting for a possible combination of different transport mechanisms. The energy balance does not contain any evaporation term since it is assumed that evaporation only occurs at the food surface which is actually exposed to the drying air.

The non-isothermal turbulent flow of air within the drying chamber has been modeled by means of the well-known

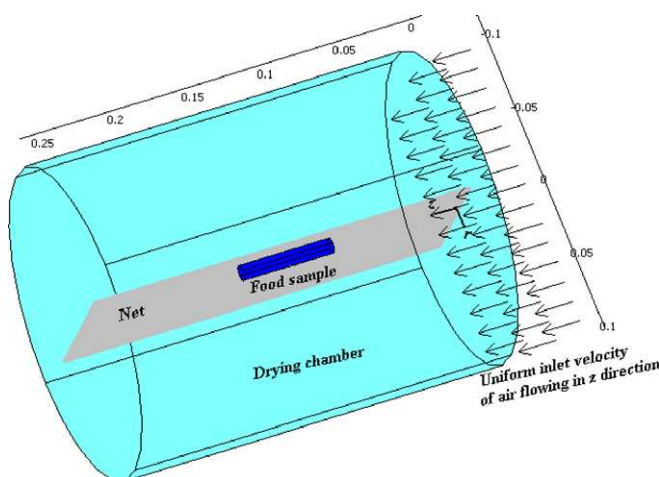


Fig. 1. Schematic representation of the drying chamber.

k - ε model (Verboven et al., 1997; Verboven et al., 2001) that is based on two additional semi-empirical transport equations for the variables k and ε , i.e. the turbulent kinetic energy and the turbulent energy dissipation rate, respectively. The unsteady-state momentum balance coupled to the continuity equation lead to (Bird et al., 1960; Verboven et al., 2001)

$$\partial \rho_a / \partial t + \nabla \cdot \rho_a \underline{u} = 0 \quad (3)$$

$$\rho_a \partial \underline{u} / \partial t + \rho_a \underline{u} \cdot \nabla \underline{u} = \nabla \cdot [-p \underline{I} + (\eta_a + \eta_t)(\nabla \underline{u} + (\nabla \underline{u})^T - (2/3)(\nabla \cdot \underline{u})\underline{I})] \quad (4)$$

$$\rho_a \partial k / \partial t + \rho_a \underline{u} \cdot \nabla k = \nabla \cdot [(\eta_a + \eta_t / \sigma_k)(\nabla k)] + \eta_t P(\underline{u}) - (2\rho_a k / 3)(\nabla \cdot \underline{u}) - \rho_a \varepsilon \quad (5)$$

$$\rho_a \partial \varepsilon / \partial t + \rho_a \underline{u} \cdot \nabla \varepsilon = \nabla \cdot [(\eta_a + \eta_t / \sigma_\varepsilon)(\nabla \varepsilon)] + (c_{1\varepsilon} \varepsilon / k)[\eta_t P(\underline{u}) - (2\rho_a k / 3)(\nabla \cdot \underline{u})] - c_{2\varepsilon} \rho_a \varepsilon^2 / k \quad (6)$$

where ρ_a is the air density and η_a is its viscosity, both expressed in terms of the local values of temperature and of water content; p is the pressure within the drying chamber; $\underline{u}$ is the velocity vector; c_μ , σ_k , σ_ε , $c_{1\varepsilon}$ and $c_{2\varepsilon}$ are constants whose value depends on the k - ε turbulence model used. In the present analysis, the standard k - ε model has been adopted (Verboven et al., 2000). The term, $P(\underline{u})$, contains the contribution of the shear stresses:

$$P(\underline{u}) = \nabla \underline{u} : (\nabla \underline{u} + (\nabla \underline{u})^T) - (2/3)(\nabla \cdot \underline{u})^2 \quad (7)$$

The following definition for η_t , i.e. the turbulent viscosity, in Eqs. (4)–(6), holds

$$\eta_t = \rho_a c_\mu k^2 / \varepsilon \quad (8)$$

The energy balance in the drying air, accounting for both convective and conductive contributions, leads to (Bird et al., 1960; Welty et al., 2001)

$$\rho_a C_{pa} \partial T_2 / \partial t - \nabla \cdot (k_a \nabla T_2) + \rho_a C_{pa} \underline{u} \cdot \nabla T_2 = 0 \quad (9)$$

where T_2 is the air temperature; C_{pa} is its specific heat; and k_a is the thermal conductivity.

The water mass balance in the drying air, accounting for both convective and diffusive contributions, leads to (Bird et al., 1960; Welty et al., 2001)

$$\partial C_2 / \partial t + \nabla \cdot (-D_a \nabla C_2) + \underline{u} \cdot \nabla C_2 = 0 \quad (10)$$

where C_2 is water concentration in the air and D_a is the diffusion coefficient of water in air. Since physical and transport properties of both air and food are expressed in terms of the local values of temperature and moisture content, Eqs. (1)–(10) represent a system of unsteady, non-linear partial differential equations which can only be solved by means of a numerical method.

A set of initial conditions, typically prevailing in an industrial drying process (Table 1), is necessary to perform the numerical simulations. As far as the air is concerned, it was assumed that: the water concentration, C_{20} , was equal

Table 1
Initial conditions used in the present theoretical model

Initial condition	Description	Value
C_{20}	Water concentration in air	0.729 mol/m ³
T_{20}	Air temperature	318 K
p_0	Pressure in the drying chamber	1 atm
$\underline{u}$	Air velocity	$\underline{0}$
T_0	Food temperature	289 K
C_0	Moisture content in food	44860 mol/m ³ = 0.85 kg H ₂ O/kg _{db} kg _{wb} = 5.67 kg H ₂ O/kg _{db}

to 0.729 mol/m³ (corresponding to a relative humidity, U_{r0} , of 20%); the temperature, T_{20} , took the value 318 K; and the pressure in the drying chamber, p_0 , was 1 atm. It was also assumed that the air was stationary (i.e. $\underline{u} = \underline{0}$). The initial values of the food temperature, T_0 , and its moisture content, C_0 , were set equal to 289 K and 44860 mol/m³, respectively. The adopted C_0 value refers to an average initial moisture content, typical of fresh carrots corresponding to 0.85 kg H₂O/kg_{wb} (on wet basis) or to 5.67 kg H₂O/kg_{db} (on dry basis). The boundary conditions are reported, for the sake of brevity, in Fig. 2 where each different boundary is identified by an integer ranging from 1 to 9. At the food–air interface (boundaries 7–9), where no accumulation occurs, the continuity of both heat and water fluxes is imposed. In particular, the heat transported by convection and conduction from air to food is partly used to raise the sample temperature by conduction and partly to allow water evaporation described by considering the latent heat of vaporization (λ). Similarly, a balance also applies between the diffusive flux of liquid water coming from the core of the sample, and the flux of vapor leaving the food surface and transported into the drying air. Moreover the thermodynamic equilibrium at the air–food interface can be expressed as (Smith et al., 1987)

$$\gamma_w x_w f_w^0 = \hat{\phi}_w y_w p \quad (11)$$

where γ_w , the activity coefficient of water and f_w^0 , the fugacity of water, refer to the liquid phase, $\hat{\phi}_w$, the fugacity coefficient of water, refers – instead – to the vapor phase; x_w and y_w are the mole fractions of water in food and in air, respectively; and p is the pressure within the drying chamber. At low pressures, the vapor phase usually approximates to ideal gas behavior and Eq. (11) can be simplified as

$$\gamma_w x_w P_w^{\text{sat}} = y_w p \quad (12)$$

where P_w^{sat} is the vapor pressure of water at the interface temperature. It may be noted that in hygroscopic materials, like most of the foods, the parameter γ_w accounts for the effects of physically bound water; it is, therefore, usually expressed (as in the present study) as a function of both food moisture content and its temperature (Datta, 2007a, Part II; Ruiz-López et al., 2004). Once this relationship be-

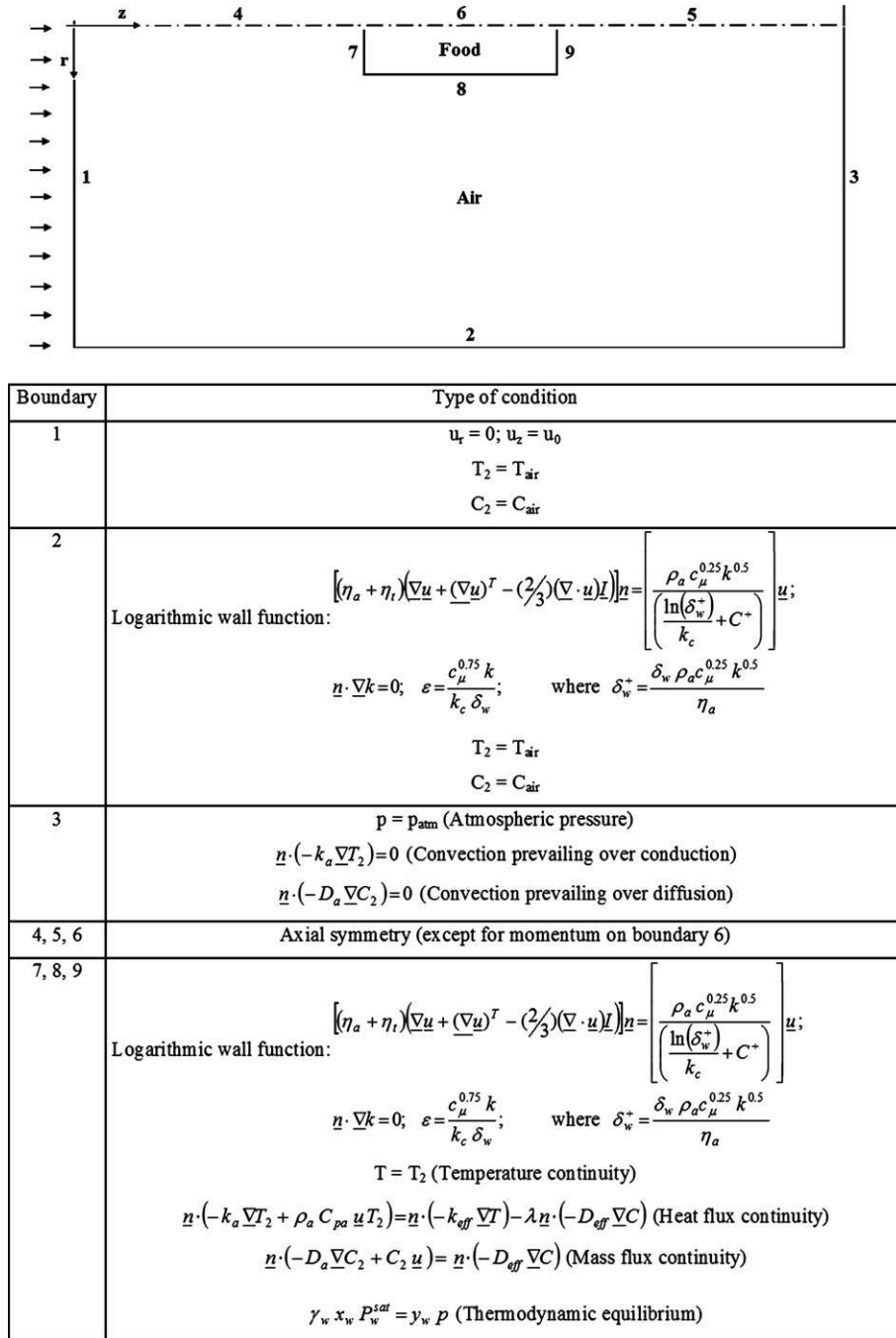


Fig. 2. Boundary conditions used to formulate the theoretical model.

tween activity coefficient and moisture content, and temperature is known for the particular food under examination, Eq. (12) permits calculating the mole fraction of water in the vapor phase and, therefore, the value of C_{2i} (i.e. water concentration evaluated in gaseous phase at the food/air interface).

On boundary 2, it is assumed that air temperature and its moisture concentration are equal to the values measured at the drier inlet. These two conditions are valid under the assumption that the temperature and concentration profiles are confined to two very thin regions which develop

close to the food–air interface. The validity of the above assumptions will be considered while discussing the simulations results. At the drier outlet (boundary 3), conduction and diffusion can be neglected in favour of convection which dominates (Danckwerts' conditions). Finally, the boundary conditions for momentum balance at the solid surfaces are expressed in terms of a two-velocity scale wall function, as reported by [Lacasse et al. \(2004\)](#).

The above system of unsteady, non-linear PDEs has been solved by Finite Elements Method using a commercial package, i.e. Comsol Multiphysics 3.3. Both food and air

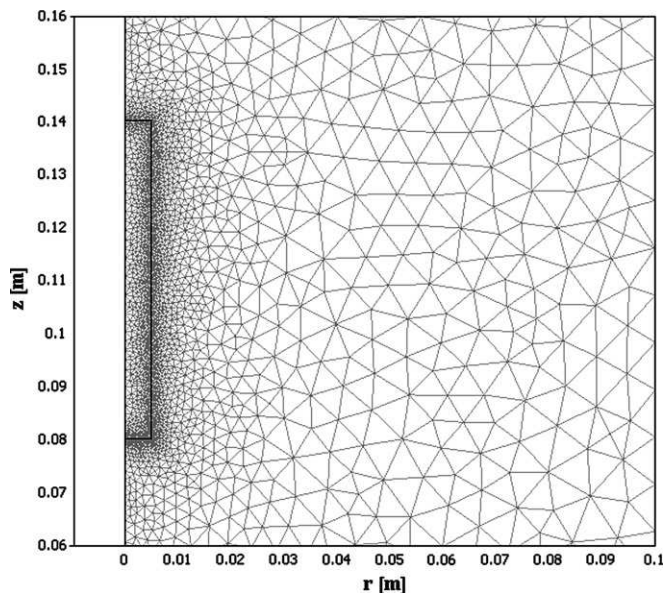


Fig. 3. Discretization of food and air domains into triangular finite elements (detail of the mesh).

domains were discretized into a total number of 5470 triangular finite elements leading to about 57,500 degrees of freedom. In particular, the mesh consisted of 1576 and 3894 elements within the food and air domains, respectively. Fig. 3 shows the details of the mesh considered, which provided a satisfactory spatial resolution for the system under study. It was also found that the solution was independent of the grid size even with further refinements. Lagrange finite elements of order two were chosen for the components, u_r and u_z , of air velocity vector $\underline{u}$, for the turbulent kinetic energy and the turbulent energy dissipation rate but also for water concentration and temperature in, both, air and food. A Lagrange finite element of order one was used for variable p , i.e. the pressure distribution within the drying chamber. The time-dependent problem was solved by an implicit time-stepping scheme, leading to a non-linear system of equations for each time step. Newton's method was used to solve each non-linear system of equations, whereas a direct linear system solver was adopted to solve the resulting systems of linear equations. The relative and absolute tolerances were set to 0.005 and 0.0005, respectively. On a Pentium IV computer running under Linux, a typical drying time of 5 h was simulated in about 45 min.

3. Experimental

The validity of the proposed theoretical model was ascertained by undertaking experiments using cylindrical carrots samples, dried by air of known characteristics. A standard procedure was adopted to perform the experiments which aimed to monitor the time evolution of the average moisture content of carrots and a single value of temperature measured at a point close to the surface where

the drying air impacted with the sample. Each experiment was repeated twice to ascertain the reproducibility of the results. Nevertheless, only the average value of each of the two measurements will be reported here, since the calculated standard deviations never exceeded 5%. Carrots were purchased from the local market. They were cut by a proper tool that allowed obtaining regular cylinders having a diameter of 1 ± 0.01 cm and a length of 6 ± 0.01 cm. The above dimensions were measured by a vernier caliper providing a precision of 0.02 mm. Some fragments, collected randomly from the same carrot used to perform the experiments, were used to measure the initial moisture content of each sample by an electronic moisture analyzer (HB43, Mettler Toledo). The cylindrical carrot samples were put into a lab-scale drying cell consisting of a cylinder having a diameter 15 cm and a length 18 cm, equipped with a stainless steel wide-mesh net where each sample was placed. To determine the time evolution of the average moisture content, weight losses were measured at given time intervals by a precision balance, Sartorius CP225D-OCE. To follow the time evolution of local temperature within the sample, a capillary thermocouple (PTFE, K, PK, RS components) was inserted 0.8 mm deep inside the carrot sample, 1 cm from the impact surface. The thermocouple, having a diameter of 0.6 mm, did not interfere with the measurements, and it was connected to a two-channel digital thermometer (KRS52 dual, RS components). Drying air was fed to the cell by an electric fan that allowed selecting air inlet flow rate, its temperature and the relative humidity. The drying air entered in the axial direction and flowed parallel to the main axis of the food sample. Air characteristics, i.e. velocity, temperature and relative humidity were continuously monitored at the inlet section of the drying cell by an Atmos anemometer. Different conditions were chosen to test the theoretical model proposed. However, the experimental results obtained at three different air temperatures (308 K, 311 K and 320 K), two different values of relative humidity (16% and 22 %) and two different air velocities (1 m/s and 1.5 m/s) will be discussed in this paper. It may be noted that the operating conditions selected were such that shrinkage effects, not accounted for by the present model, occurred only to a limited extent and in the very last part of each experiment.

4. Results and discussion

The analysis of the transport phenomena presented here is based on the examination of the time evolutions of averaged variables, i.e. the dimensionless moisture content and the temperature on each exposed surface, and water concentration distributions. Simulations at different drying conditions were performed to show how air characteristics (dry bulb temperature, relative humidity and inlet velocity) influenced the drying process and, in particular, the drying rate. It should be remarked that turbulent conditions prevailed in the drying chamber in the range of operating vari-

ables tested, since the Reynolds numbers encountered were well in excess of 10^4 .

Figs. 4–6 show the time evolutions of dimensionless average moisture content, ϕ , with air inlet temperature, relative humidity and air velocity, respectively, as parameters. ϕ is defined as

$$\phi(t) = \frac{\bar{X}(t) - X_e}{\bar{X}_0 - X_e} \quad (13)$$

where $\bar{X}(t)$ is the moisture content (on a dry basis), averaged over the sample volume and evaluated as a function of drying time; $\bar{X}_0$ is the average initial moisture content; and X_e is the equilibrium moisture content as defined by Ruiz-López et al. (2004). An increase in air inlet temperature from 308 K to 328 K (Fig. 4) results in an increase in wet bulb temperature from 292 K to about 305 K (Perry and Green, 1984) that drives the system away from equilib-

rium conditions determining an initial delay in the start-up of drying. At the very beginning of the process all the curves overlap showing an almost identical slope; then, they part evidencing a drying rate (proportional to the slopes of ϕ vs. time curves) that is higher and higher as dry bulb temperature increases. The increase in wet bulb temperature may potentially cause an initial humidification of the exposed surfaces, especially at higher values of air relative humidity. This phenomenon is caused by the condensation of water from the drying air on the cold food surfaces and continues until the surface temperature is lower than wet bulb temperature. Once the wet bulb temperature is attained, evaporation of free water dominates over vapor condensation and the food actually begins to dry. Moreover, at a given relative humidity, an increase in the dry bulb temperature results in an increase in the moisture content of the air which causes a significant reduction in

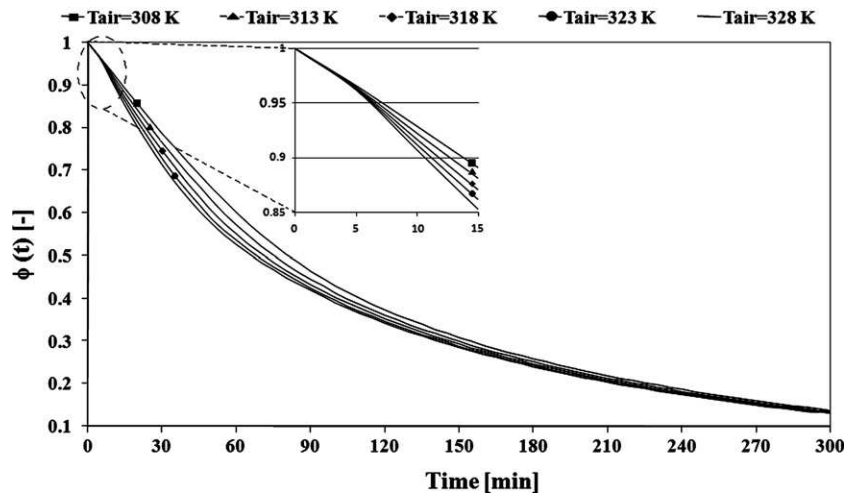


Fig. 4. Time evolution of dimensionless average moisture content with inlet air temperature as the parameter ($T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg $_{wb}$, $u_0 = 1.5$ m/s, $U_r = 20\%$).

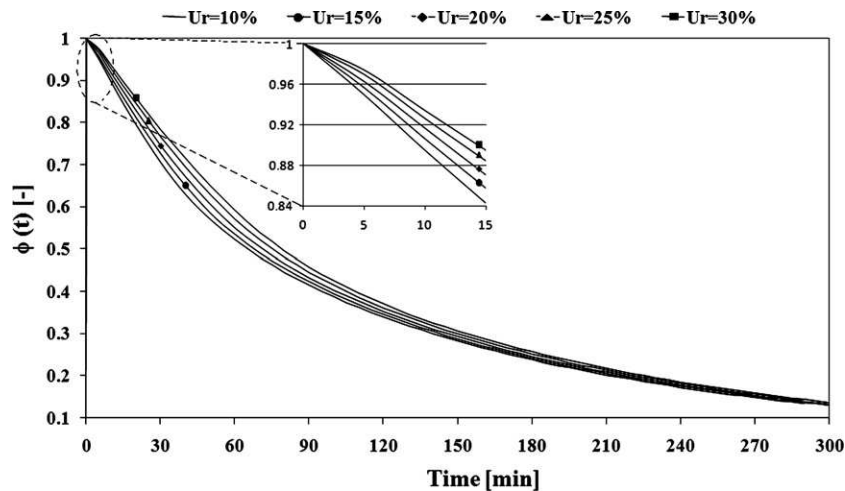


Fig. 5. Time evolution of dimensionless average moisture content with inlet air relative humidity as the parameter ($T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg $_{wb}$, $u_0 = 1.5$ m/s, $T_{air} = 318$ K).

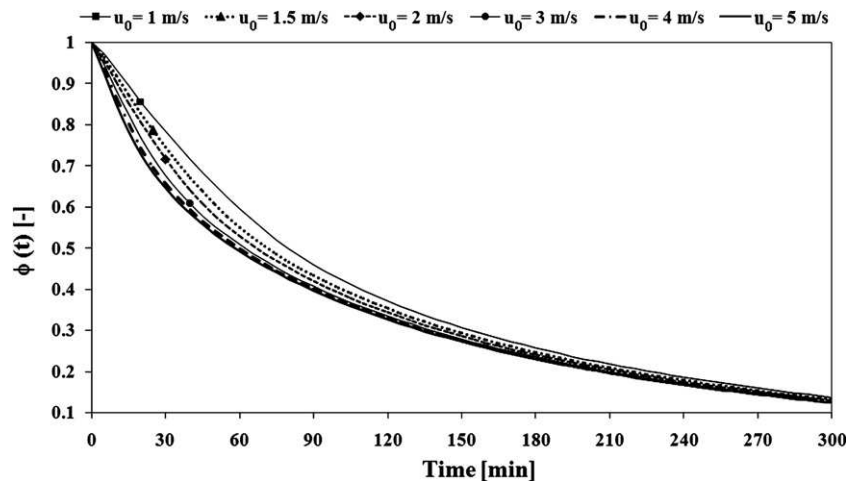


Fig. 6. Time evolution of dimensionless average moisture content with inlet air velocity as the parameter ($T_0 = 289$ K, $U_0 = 0.85$ kg $\text{H}_2\text{O}/\text{kg}_{\text{wb}}$, $T_{\text{air}} = 318$ K, $U_r = 20\%$).

the driving force to mass transfer from food to air; however, this is accompanied by an improvement in heat transfer from air to food. The observed rate of drying therefore depends on air inlet temperature during the initial stage of the process, where higher inlet temperature causes faster drying. On the other hand, the effect of air temperature on drier performance is less important in the last three hours of the process, when bound water is removed and the internal resistance to mass transport prevails and represents the limiting step. During this stage, it can be observed from Fig. 4 that the drying rate is much slower, and ϕ values and the corresponding drying rates almost coincide regardless of the inlet air temperature.

When the relative humidity of inlet air increases from 10% to 30%, an increase in, both, the wet bulb temperature by about 7.5 K, as well as its moisture content by a factor of about three, is observed. Therefore, an increase in relative humidity results in a lower drying rate and causes a

longer initial delay before drying actually commences (Fig. 5). Moreover, when free water is removed from the food, an increase in relative humidity of air from 10% to 30% significantly lowers the drying rate owing to a reduction in the driving force which promotes moisture transfer. Nevertheless, when bound water is involved in the drying process, the internal resistance to mass transfer represents the rate limiting step, and the concentration difference outside the food sample does not significantly influence the drying rate. Also, in this case, the various curves showing ϕ vs. time tend to coincide during the later stage of the drying process.

Fig. 6 shows the effect of air inlet velocity on moisture removal. An increase in air velocity corresponds to a decrease in external resistances to heat and mass transfer. This results in faster drying which can be clearly observed when free water evaporates and leaves the food, although no significant variation is observed when inlet velocity is

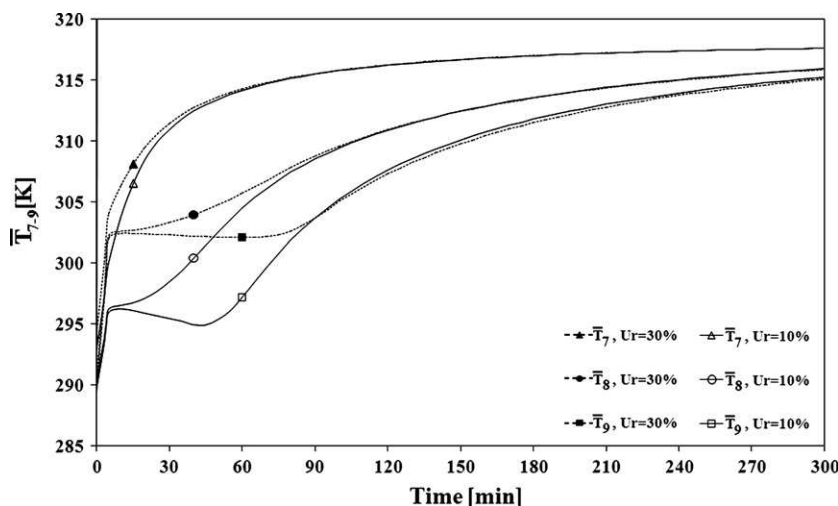


Fig. 7. Time evolution of average temperature on each food surface with inlet air relative humidity as the parameter ($T_0 = 289$ K, $U_0 = 0.85$ kg $\text{H}_2\text{O}/\text{kg}_{\text{wb}}$, $u_0 = 1.5$ m/s, $T_{\text{air}} = 318$ K).

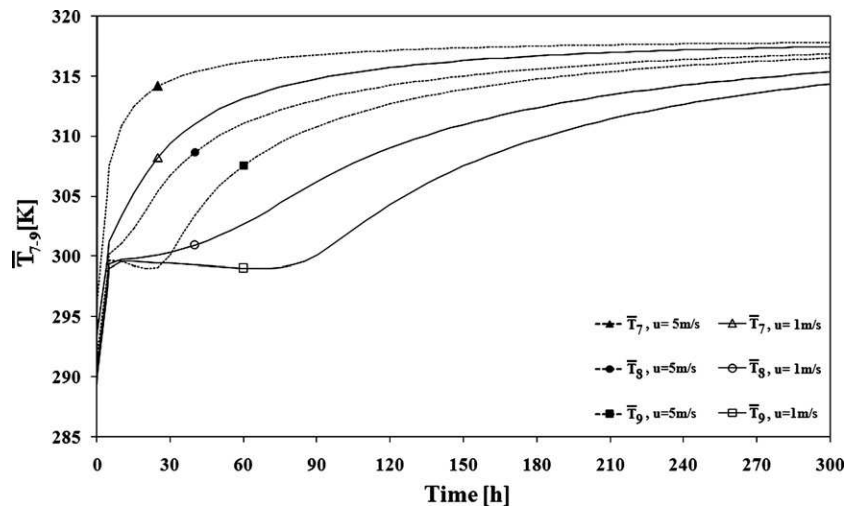


Fig. 8. Time evolution of average temperature on each food surface with inlet velocity as the parameter ($T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg $_{wb}$, $U_r = 20\%$, $T_{air} = 318$ K).

increased, for example, from 4 to 5 m/s. As soon as bound water is removed from the food, internal resistance to mass transfer predominates and air velocity has a weak influence on the process rate, since the value of effective diffusivity of water in food, dependent on both temperature and moisture content, controls the drying rate. The present model confirms that changes in dry bulb temperature, inlet humidity and inlet velocity influence the drying rates when free water is involved in the transfer process. On the other hand, when bound water is removed from the food and internal resistances control the process rate, the operating variables have little effect on the system behavior. The proposed model may represent a powerful tool to analyze the

behavior of industrial driers over a wide range of process and fluid-dynamic conditions. It may also be used to find an “optimal” set of operating conditions aimed at improving the final product quality.

The time evolution of food sample temperature, averaged on each surface exposed to the drying air, i.e. boundaries 7–9, are shown in Figs. 7 and 8. The plots were obtained at two different values of relative humidity and inlet velocity, respectively. The air–impact food surface (boundary 7) is heated more rapidly than the other surfaces and its temperature rises quickly above the wet bulb temperature. This behavior is definitely ascribed to the flow conditions which have a strong influence on the external

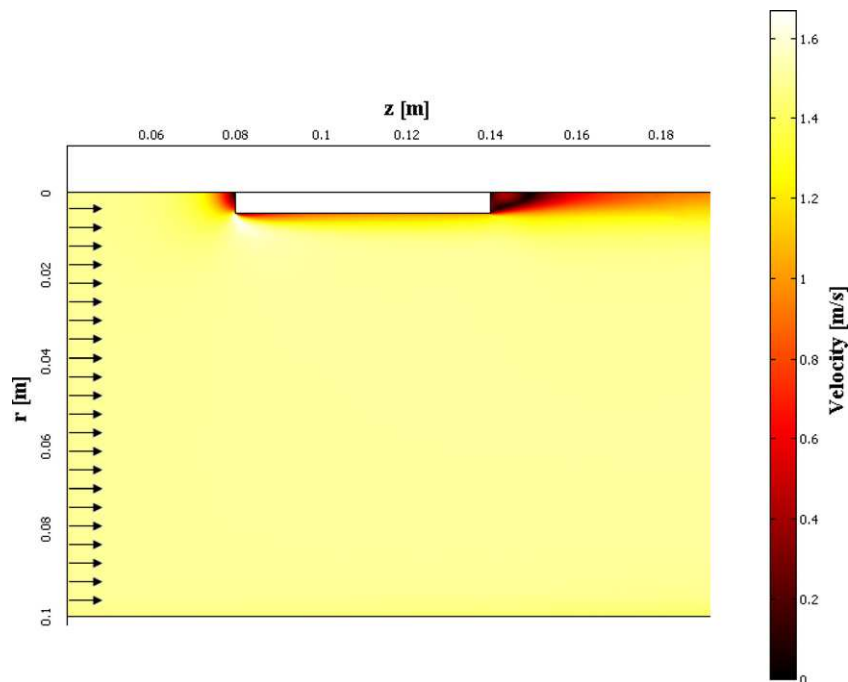


Fig. 9. Air velocity field developing close to the food sample ($t = 60$ min, $T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg $_{wb}$, $u_0 = 1.5$ m/s, $T_{air} = 318$ K, $U_r = 20\%$).

Table 2
Comparison among wet bulb temperatures as predicted by the proposed model and those ones obtained by the psychrometric chart

Dry bulb temperature [K]	Air relative humidity (%)	Wet bulb temperature [K] (psychrometric chart)	Wet bulb temperature [K] (predicted by the model)	Relative error (%)
318.15	20	298.35	299.75	0.47
308.15	20	292.05	293.55	0.51
313.15	20	295.15	296.65	0.51
323.15	20	301.55	302.85	0.43
328.15	20	304.75	305.95	0.39
318.15	10	294.25	295.25	0.34
318.15	15	296.35	298.05	0.57
318.15	25	300.15	301.35	0.40
318.15	30	301.85	302.55	0.23

resistances to both heat and mass transfer. At the impact surface: (1) the heat transferred from air provides both the latent heat of vaporization necessary for water evaporation and the sensible heat necessary to increase the temperature of the dry matter by conduction and (2) the drying rate is immediately very high and it involves evaporation of free water which is transported into the air. A completely different behavior is observed at the opposite end, i.e. on boundary 9, where a rather wide “segregation” region forms due to inefficient air circulation, as shown by the velocity field shown in Fig. 9. The time evolutions of average temperature corresponding to the “rear” end of the food surface show that, at the very beginning of drying process, the heat transported from air increases the food

temperature up to the wet bulb temperature; later on, the average temperature is nearly constant. During this stage, the duration of which depends on the operating conditions (shorter when air relative humidity is low or inlet velocity is high), the heat transferred from air provides only the latent heat of evaporation to the free water. When bound water starts to get eliminated, the heat transported by convection and conduction from air to food is partially used to raise the sample temperature by conduction and partially to allow water evaporation. The average temperature continuously increases up to a plateau value, with a slope that depends on the values of air velocity and relative humidity. The behavior of the food surface parallel to the direction of air flow (boundary 8) can be considered to be “intermediate” between that of the other two boundaries. Temperature profiles do exhibit a nearly constant value whose duration is, however, shorter, when air relative humidity is equal to 10%, or tends to vanish as soon as the air velocity reaches its highest value, thus determining a significant decrease in external transport resistances. A comparison between the values of wet bulb temperature predicted by the model and that obtained from the psychrometric chart (Perry and Green, 1984), under several conditions, is given in Table 2; a very good agreement is evident with relative errors being at most 0.5%.

The proposed model can also describe the temperature and spatial moisture profiles in both air and food at any time. This is highly relevant from the food safety point of view, inasmuch as it allows detection of the regions within the food core where high values of moisture content might

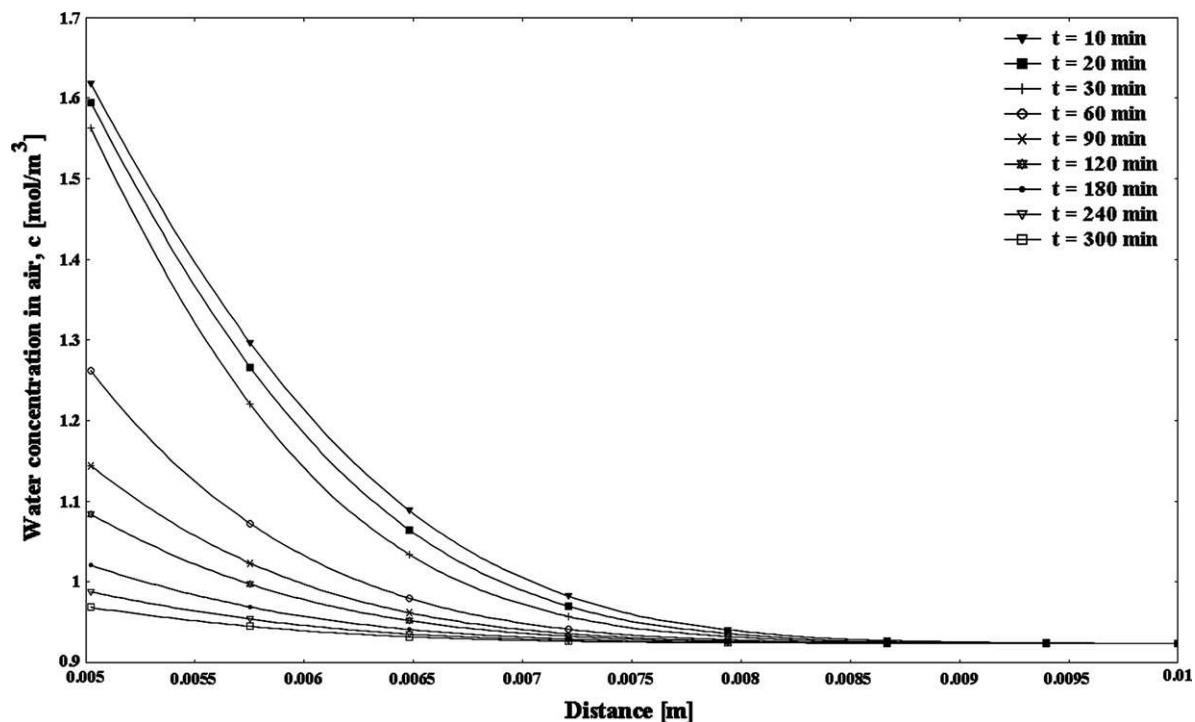


Fig. 10. Detail of water concentration profiles developing in air in the radial direction at different drying times (axial position: $z = 0.125$ m (half length of the drier), $T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg w_{wb} , $u_0 = 1.5$ m/s, $T_{air} = 318$ K, $U_r = 20\%$).

determine microbial spoilage. Fig. 10 shows the water concentration in the air, as a function of the radial distance from the food sample at different drying times. It can be observed that a concentration gradient develops in a very thin region close to the food surface as predicted, for instance, by the boundary layer theory (Schlichting, 1960). Fig. 10 confirms that the above boundary conditions referred to water concentration at boundary 2 (see Fig. 2) may be considered as a true representation of behavior of a real system. Concentration gradients, in fact, vanish at

a very short distance, equal to about 3 mm, from the food surface in comparison to the drier radius of 10 cm. It can be also noted that the external resistances to mass transfer tend to decrease as drying proceeds; this is further evidence that internal resistance to water transport actually controls the final stage of the process. Similar conclusions can also be drawn in respect of air temperature profiles (data not presented). Fig. 11 shows the moisture content profiles developing within the food sample in the radial direction at different times. It can be observed that the internal mass

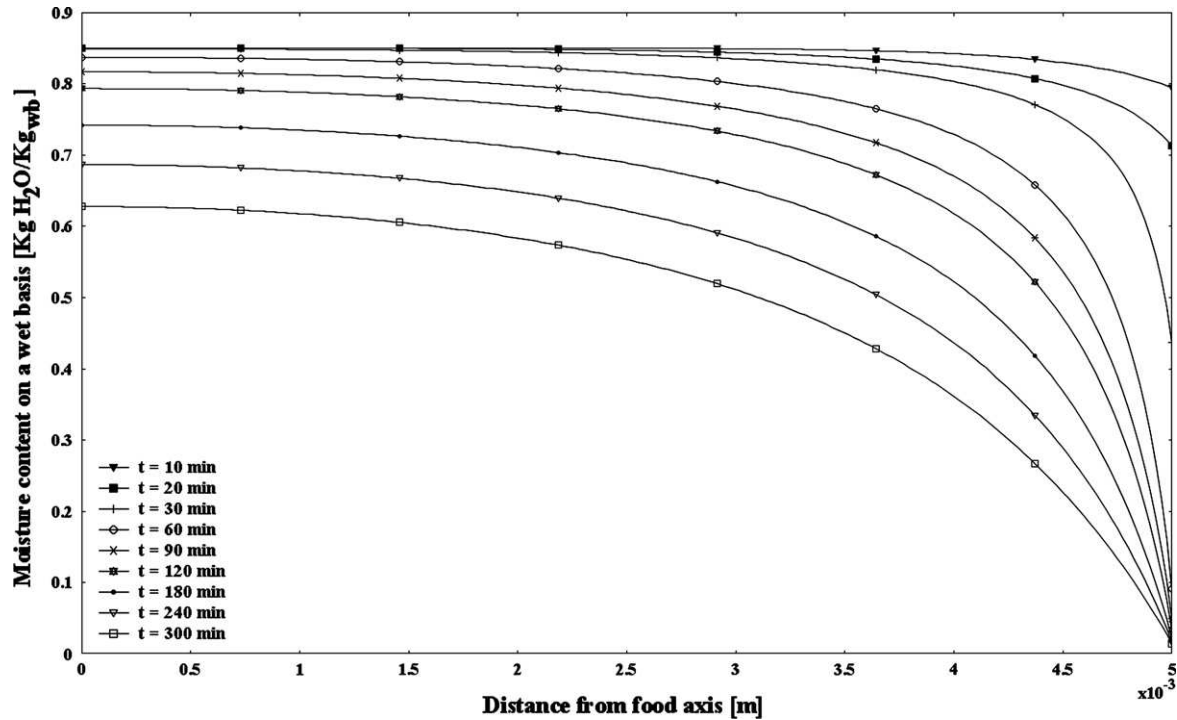


Fig. 11. Moisture content profiles developing within the food in radial direction at different drying times (axial position: $z = 0.11$ m (half length of the food), $T_0 = 289$ K, $U_0 = 0.85$ kg H_2O /kg_{wb}, $u_0 = 1.5$ m/s, $T_{air} = 318$ K, $U_r = 20\%$).

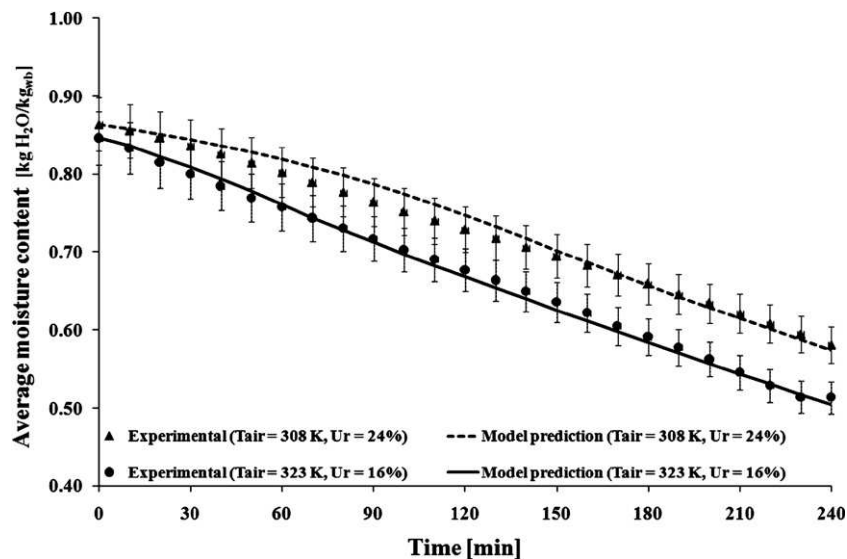


Fig. 12. Comparison between model predictions and experimental data – time evolution of average moisture content on a wet basis ($u_0 = 1$ m/s).

transfer resistance plays a progressively stronger role as drying proceeds. The moisture content gradient developing in radial direction is, in fact, more pronounced as process time advances; this is also substantiated by the greater differences between water concentration at the sample core and that at the surface exposed to air.

5. Model validation

In addition to the numerical simulations described in the previous section, other simulations were performed with the specific aim of verifying the model validity, by checking the agreement between experimental results, obtained under different fluid-dynamic and operating conditions and model predictions. Model validation has been carried out for the case of air-drying performed on cylindrical-shaped carrot samples. Fig. 12 shows the comparison between experimental and predicted average moisture content during two drying tests performed at the same value of air velocity (1 m/s) but changing inlet air temperature and its relative humidity. In the milder condition, the dry bulb temperature was maintained at 308 K and air relative humidity was equal to 24%; in the more severe case, the air temperature was 323 K and relative humidity was 16%. Experimental data and model predictions show remarkable agreement, with relative errors never exceeding 3% in the second case, and 1.7% in the second case. Fig. 13 shows that the proposed model is also capable of giving very good predictions of the actual time evolution of food temperature. As mentioned earlier, the food temperature was experimentally measured by a capillary thermocouple inserted in a specific point located 1 cm away from the impact surface at a depth of 0.8 mm. In this case, the effect of air inlet velocity on drying behavior has been taken into account. The agreement between experimental data and

model predictions is rather good with relative errors never exceeding 1%. The proposed model therefore gives a good representation of the behavior of the real system, since it is capable of reproducing experimental data not only at the very beginning of drying process, but also over the entire time duration.

6. Conclusions

The transport phenomena involved in food drying have been analyzed. A general predictive model, i.e. one not based on any semi-empirical correlation for estimating heat and mass fluxes at food–air interface, has been formulated. The model, which also accounts for momentum transport of air flowing around the food sample, was capable of describing the real drier behavior over a wide range of process and fluid-dynamic conditions. The proposed model is particularly useful for those situations for which, either semi-empirical correlations are not currently available (i.e. complex food geometries), or operating conditions are changed during drying process. Moreover, the model can also be used to determine which set of operating conditions would enhance the quality and the safety of the final product. It was observed that air properties (dry bulb temperature, relative humidity, inlet velocity) play a significant role only during the initial stage of the drying process, i.e. when external resistances to heat and mass transfer control the drying rate. On the other hand, when bound water is removed from the food and internal resistances control the process rate, the operating variables have insignificant effect on the system behavior. The proposed model also predicted the spatial moisture profiles at all times, thus allowing detection of the regions within the food core, where high values of moisture content can promote microbial spoilage. Simulations results were found to be in good

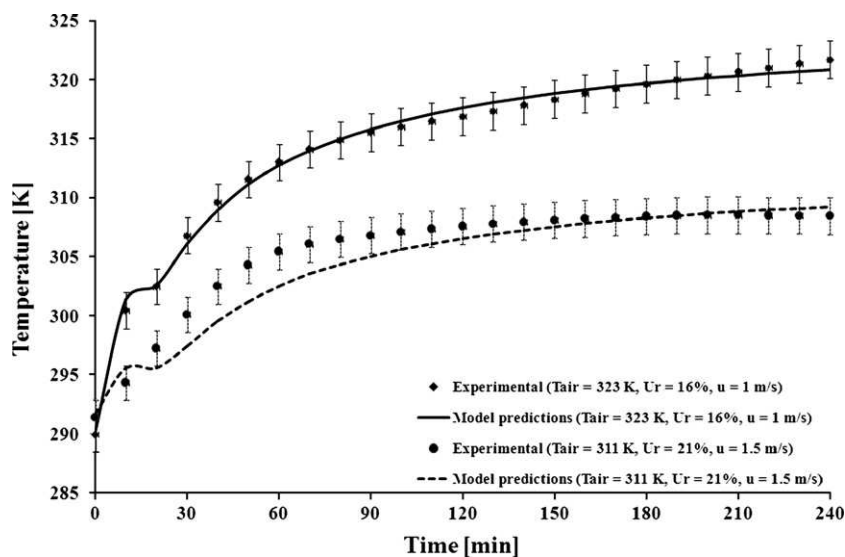


Fig. 13. Comparison between model predictions and experimental data – time evolution of temperature 1 cm from the impact surface at a depth of 0.8 mm.

agreement with experimental data in terms of time evolutions of food moisture content and temperature. The model can be improved by taking into account phenomena such as product shrinkage and transport of water vapor, by diffusion, within the dehydrated material, which can influence the drying rate and, therefore, the quality of final product.

References

- Aversa, M., Curcio, S., Calabrò, V., Iorio, G., 2007. An analysis of the transport phenomena occurring during food drying process. *Journal of Food Engineering* 78 (3), 922–933.
- Bird, R.B., Stewart, W.E., Lightfoot, E.N., 1960. *Transport Phenomena*. John Wiley & Sons, London, UK.
- Datta, A.K., 2007a. Porous media approaches to studying simultaneous heat and mass transfer in food processes. Part I: problem formulations. *Journal of Food Engineering* 80, 80–95.
- Datta, A.K., 2007b. Porous media approaches to studying simultaneous heat and mass transfer in food processes. Part II: property data and representative results. *Journal of Food Engineering* 80, 96–110.
- Hernández, J.A., Pavón, G., García, M.A., 2000. Analytical solution of mass transfer equation considering shrinkage for modeling food-drying kinetics. *Journal of Food Engineering* 45, 1–10.
- Kalbasi, M., Mehraban, M.R., 2000. The effect of surface water vapor flux on drying of potato. *J. Trans. IChemE (Part C)* 78, 204–206.
- Kondjoyan, A., Boisson, H.C., 1997. Comparison of calculated and experimental heat transfer coefficients at the surface of circular cylinders placed in a turbulent cross-flow of air. *Journal of Food Engineering* 34, 123–143.
- Lacasse, D., Turgeon, E., Pelletier, D., 2004. On the judicious use of the K-Epsilon model, wall functions and adaptivity. *International Journal of Thermal Sciences* 43 (10), 925–938.
- Lewicki, P.P., 2004. Water as determinant of food engineering properties. A review. *Journal of Food engineering* 61, 483–495.
- May, B.K., Perré, P., 2002. Importance of considering exchange surface area reduction to exhibit a constant drying flux period in food stuffs. *Journal of Food Engineering* 54, 271–282.
- Migliori, M., Gabriele, D., de Cindio, B., Pollini, M.C., 2005. Modelling of high quality pasta drying: mathematical model and validation. *Journal of Food Engineering* 69, 387–397.
- Mujumdar Arun, S., 2006. *Handbook of Industrial Drying*, third ed. Taylor and Francis, ISBN 1574446681.
- Panagiotou, N.M., Krokida, M.K., Maroulis, Z.B., Saravacos, G.D., 2004. Moisture diffusivity: literature data compilation for foodstuff. *International Journal of Food Properties* 7 (2), 273–299.
- Perry, R.H., Green, D., 1984. *Perry's Chemical Engineers' Handbook*. McGraw-Hill, New York.
- Rovedo, C.O., Suarez, C., Viollaz, P.E., 1995. Drying of foods: evaluation of a drying model. *Journal of Food Engineering* 26, 1–12.
- Ruiz-López, I.I., Córdova, A.V., Rodríguez-Jimenes, G.C., García-Alvarado, M.A., 2004. Moisture and temperature evolution during food drying: effect of variable properties. *Journal of Food Engineering* 63, 117–124.
- Saravacos, G.D., Maroulis, Z.B., 2001. *Transport Properties of Foods*. Marcel Dekker Inc., New York.
- Schlichting, H., 1960. *Boundary Layer Theory*. McGraw-Hill Book Company, New York.
- Smith, J.M., Van Ness, H.C., Abbott, M.M., 1987. *Chemical Engineering Thermodynamics*, fourth ed. McGraw-Hill, New York.
- Thorvaldsson, K., Janestad, H., 1999. A model for simultaneous heat, water and vapour diffusion. *Journal of Food Engineering* 40, 167–172.
- Verboven, P., Nicolai, B.M., Scheerlinck, N., De Baerdemaeker, J., 1997. The local surface heat transfer coefficient in thermal food process calculations: a CFD approach. *Journal of Food Engineering* 33, 15–35.
- Verboven, P., Scheerlinck, N., De Baerdemaeker, J., Nicolai, B.M., 2000. Computational fluid dynamics modelling and validation of the isothermal air flow in a forced convection oven. *Journal of Food Engineering* 43, 41–53.
- Verboven, P., Scheerlinck, N., De Baerdemaeker, J., Nicolai, B.M., 2001. Sensitivity of the food centre temperature with respect to the air velocity and the turbulence kinetic energy. *Journal of Food Engineering* 48, 53–60.
- Viollaz, P.E., Rovedo, C.O., 2002. A drying model for three-dimensional shrinking bodies. *Journal of Food Engineering* 52, 149–153.
- Wang, N., Brennan, J.G., 1995. A mathematical model of simultaneous heat and moisture transfer during drying of potato. *Journal of Food Engineering* 24, 47–60.
- Wu, Y., Irudayaraj, J., 1996. Analysis of heat, mass and pressure transfer in starch based food systems. *Journal of Food Engineering* 29, 399–414.
- Welty, J.R., Wicks, C.E., Wilson, R.E., Rorrer, G., 2001. *Fundamentals of Momentum, Heat and Mass Transfer*, fourth ed. John Wiley & Sons, New York.